

NetBox Installation Final

Overview

This documentation describes how to install NetBox and configure NetBox with Let's encrypt and SAML for SSO.

I remove the step for self-signed certificate, Let's Encrypt is the replacement.

WARNING: if you copy and paste be careful of white spaces in configuration files

Prerequisite

- Ubuntu-22.04
- All updates/upgrade completed (sudo apt update && upgrade)
- Nginx (apt install nginx)
- Hosts file is configured
- Hostname file is configured
- Date/Time corrected (sudo timedatectl set-timezone America/Chicago)
- Static IP Address
- Install locate (apt install mlocate)
- Net-tools (apt install net-tools)

PostgreSQL Database Installation

```
sudo apt install -y postgresql
```

Check Version

```
psql -V
```

Database Creation

```
sudo -u postgres psql
```

Database Create

```
CREATE DATABASE netbox;  
CREATE USER netbox WITH PASSWORD 'J5brHrAXFLQSi0K';  
ALTER DATABASE netbox OWNER TO netbox;
```

```
-- the next two commands are needed on PostgreSQL 15 and later
\connect netbox;
GRANT CREATE ON SCHEMA public TO netbox;
```

Verify Service Status

```
psql --username netbox --password --host localhost netbox
```

Redis Installation

```
sudo apt install -y redis-server
```

Check Version

```
redis-server -v
```

Verify Service Status

```
redis-cli ping
```

NetBox Installation

```
sudo apt install -y python3 python3-pip python3-venv python3-dev build-essential libxml2-dev libxslt1-dev libffi-  
dev libpq-dev libssl-dev zlib1g-dev
```

Check Version

```
python3 -V
```

Download NetBox

```
sudo wget https://github.com/netbox-community/netbox/archive/refs/tags/v3.6.5.tar.gz
```

Extract package

```
sudo tar -xzf v3.6.5.tar.gz -C /opt
```

```
sudo ln -s /opt/netbox-3.6.5/ /opt/netbox
```

Create System user

```
sudo adduser --system --group netbox
sudo chown --recursive netbox /opt/netbox/netbox/media/
sudo chown --recursive netbox /opt/netbox/netbox/reports/
sudo chown --recursive netbox /opt/netbox/netbox/scripts/
```

Configuration

```
cd /opt/netbox/netbox/netbox/
sudo cp configuration_example.py configuration.py
```

Generate a key for SECRET_KEY

```
python3 ../generate_secret_key.py
```

NetBox Configuration file (configuration.py)

ALLOWED_HOST will be configured to **localhost** till the certificates are made from CertBot below.

```
ALLOWED_HOSTS = ['localhost']
DATABASE = {
    'NAME': 'netbox',          # Database name
    'USER': 'netbox',          # PostgreSQL username
    'PASSWORD': 'J5brHrAXFLQSif0K', # PostgreSQL password
    'HOST': 'localhost',      # Database server
    'PORT': '',                # Database port (leave blank for default)
    'CONN_MAX_AGE': 300,      # Max database connection age (seconds)
}
```

```
REDIS = {
    'tasks': {
        'HOST': 'localhost',   # Redis server
        'PORT': 6379,          # Redis port
        'PASSWORD': '',        # Redis password (optional)
        'DATABASE': 0,         # Database ID
        'SSL': False,          # Use SSL (optional)
    },
}
```

```
'caching': {
    'HOST': 'localhost',
    'PORT': 6379,
    'PASSWORD': '',
    'DATABASE': 1,          # Unique ID for second database
    'SSL': False,
}
```

```
SECRET_KEY = '!3#qSHaL&(7mw38.....j^mKZG@$Ax'
```

Logging configuration

```
sudo mkdir /var/log/netbox
sudo touch /var/log/netbox/netbox.log
sudo chown -R netbox.netbox /var/log/netbox
```

Add this section to NetBox configuration.py file

NOTE: If you copy and Paste ensure there are no white spaces.

```
LOGGING = {
    'version': 1,
    'disable_existing_loggers': False,
    'formatters': {
        'normal': {
            'format': '%(asctime)s %(name)s %(levelname)s: %(message)s'
        },
    },
    'handlers': {
        'file': {
            'level': 'DEBUG',
            'class': 'logging.handlers.WatchedFileHandler',
            'filename': '/var/log/netbox/netbox.log',
            'formatter': 'normal',
        },
    },
    'loggers': {
```

```
'django': {
    'handlers': ['file'],
    'level': 'INFO',
},
'netbox': {
    'handlers': ['file'],
    'level': 'INFO',
},
}
```

Remote File Storage

This setting has been disabled in configuration.py file

```
# 'ENGINE': 'django.db.backends.postgresql', # Database engine
```

Upgrade Script

NOTE: This depends on the version of Python installed.

```
/opt/netbox/netbox/netbox
```

Execute upgrade script

```
sudo PYTHON=/usr/bin/python3.10 /opt/netbox/upgrade.sh
```

Create a Super User

```
source /opt/netbox/venv/bin/activate
```

```
cd /opt/netbox/netbox
python3 manage.py createsuperuser
```

Schedule the Housekeeping Task

```
sudo ln -s /opt/netbox/contrib/netbox-housekeeping.sh /etc/cron.daily/netbox-housekeeping
```

Test the Application

```
python3 manage.py runserver 0.0.0.0:8000 --insecure
```

Warning: If the test service does not run, or does not complete checks that show "OK", something has gone wrong. Call for help. Do not proceed with the rest of this guide until the installation has been corrected.

NetBox Service

Gunicorn

```
sudo cp /opt/netbox/contrib/gunicorn.py /opt/netbox/gunicorn.py
```

Systemd setup

```
sudo cp -v /opt/netbox/contrib/*.service /etc/systemd/system/  
sudo systemctl daemon-reload
```

```
sudo systemctl start netbox netbox-rq  
sudo systemctl enable netbox netbox-rq
```

Status

```
systemctl status netbox.service
```

Create SSL Certificate with Let's Encrypt

There are a few prerequisites to use Let's Encrypts. You have to accept the ToS of Let's Encrypt to register an account.

Port 80 of the node needs to be reachable from the internet. There must be no other listener on port 80.

The requested (sub)domain needs to resolve to a public IP of the Node.

CertBot Installation command for Nginx

Nginx Configuration

Install CertBot

```
sudo apt install certbot python3-certbot-nginx
```

Create Certificates

```
sudo certbot --nginx
```

```
root@keycloak:/home/ubuntu# sudo certbot --nginx
```

```
Saving debug log to /var/log/letsencrypt/letsencrypt.log
```

```
Enter email address (used for urgent renewal and security notices)
```

```
(Enter 'c' to cancel): greg.smith@enseva.com
```

```
-----  
Please read the Terms of Service at
```

```
https://letsencrypt.org/documents/LE-SA-v1.3-September-21-2022.pdf. You must  
agree in order to register with the ACME server. Do you agree?
```

```
-----  
(Y)es/(N)o: y
```

```
-----  
Would you be willing, once your first certificate is successfully issued, to  
share your email address with the Electronic Frontier Foundation, a founding  
partner of the Let's Encrypt project and the non-profit organization that  
develops Certbot? We'd like to send you email about our work encrypting the web,  
EFF news, campaigns, and ways to support digital freedom.
```

```
-----  
(Y)es/(N)o: n
```

```
Account registered.
```

```
Please enter the domain name(s) you would like on your certificate (comma and/or  
space separated) (Enter 'c' to cancel): netbox3.hungry-howard.com
```

```
Requesting a certificate for netbox3.hungry-howard.com
```

```
Successfully received certificate.
```

```
Certificate is saved at: /etc/letsencrypt/live/netbox3.hungry-howard.com/fullchain.pem
```

```
Key is saved at: /etc/letsencrypt/live/netbox3.hungry-howard.com/privkey.pem
```

```
This certificate expires on 2024-03-13.
```

```
These files will be updated when the certificate renews.
```

```
Certbot has set up a scheduled task to automatically renew this certificate in the background.
```

```
Deploying certificate
```

```
Successfully deployed certificate for netbox3.hungry-howard.com to /etc/nginx/sites-enabled/default
```

```
Congratulations! You have successfully enabled HTTPS on https://netbox3.hungry-howard.com
```

```
-----  
If you like Certbot, please consider supporting our work by:
```

* Donating to ISRG / Let's Encrypt: <https://letsencrypt.org/donate>

* Donating to EFF: <https://eff.org/donate-le>

At this point I have been using the Default site configuration this also has SSL enable. Copy default file into a new file called netbox.

Linux magic

```
cat /etc/nginx/sites-available/default | egrep -v "^\s*(#|$)" > netbox
```

Remove default from site-enabled then enable netbox site.

```
sudo rm /etc/nginx/sites-enabled/default
```

```
sudo ln -s /etc/nginx/sites-available/netbox /etc/nginx/sites-enabled/netbox
```

Restart Service

```
systemctl restart nginx
```

Plugin install for NetBox SAML

Prepping for NetBox plugin to be enabled. Configuring the following files before restarting services.

```
sudo apt-get install pkg-config libxml2-dev libxmlsec1-dev libxmlsec1-openssl xmlsec1
```

```
source /opt/netbox/venv/bin/activate
```

```
cd /opt/netbox
```

```
pip3 install xmlsec
```

```
pip3 install django3-auth-saml2
```



```
pip3 install netbox-plugin-auth-saml2
```

Add plugin's to local requirement text file

```
sudo sh -c "echo 'django_saml2_auth' >> /opt/netbox/local_requirements.txt"
sudo sh -c "echo 'netbox-plugin-auth-saml2' >> "/opt/netbox/local_requirements.txt""
```

Netbox Plugin for SSO using SAML2

Edit file netbox configuration file.

```
vi /opt/netbox/netbox/netbox/configuration.py
```

Adjust `ALLOWED_HOST` to FQDN.

```
ALLOWED_HOSTS = ['netbox3.hungry-howard.com']
```

Zitadel XML file

Retrieve the metadata from Zitadel. Add **/saml/v2/metadata** end point on Zitadel instance. Copy the Metadata and Paste it in the file called **DCIM.xml**

```
vi /opt/netbox/DCIM.xml
```

Adjust the `configuration.py` file as shown below.

```
PLUGINS = ['django3_saml2_nbplugin']
PLUGINS_CONFIG = {
    'django3_saml2_nbplugin': {
        'AUTHENTICATION_BACKEND': 'netbox.authentication.RemoteUserBackend',
        'ASSERTION_URL': 'https://netbox.hungry-howard.com/',
        'ENTITY_ID': 'https://netbox.hungry-howard.com/',
        'METADATA_LOCAL_FILE_PATH': '/opt/netbox/DCIM.xml',
        'CUSTOM_ATTR_BACKEND': {
            'USERNAME_ATTR': "emailaddress",
            'MAIL_ATTR': "emailaddress",
            'FIRST_NAME_ATTR': "givenname",
            'LAST_NAME_ATTR': "surname",
            'ALWAYS_UPDATE_USER': True,
        }
    }
}
```

```
}  
}
```

Remote Authentication configurations

NOTE: Debugging issues add `DEBUG = False` to `DEBUG = True`

```
REMOTE_AUTH_ENABLED = True  
REMOTE_AUTH_BACKEND = 'netbox.authentication.RemoteUserBackend'  
  
REMOTE_AUTH_AUTO_CREATE_USER = True
```

```
### This is option create a button for SSO is the auto redirect is NOT configured in Nginx ###  
BANNER_LOGIN = '<a href="/api/plugins/sso/login" class="btn btn-primary btn-block">Login with SSO</a>'
```

Nginx Configuration

There are two ways to configure Nginx. For this demo I choose step **#2**. I tried to add the configurations needed but failed so I created a new file **/etc/nginx/sites-available/test** then copy and pasted **#2** configuration below to that file. Save and close. Execute this command to write over netbox site file.

```
cp /etc/nginx/sites-available/test /etc/nginx/sites-available/netbox
```

NOTE: Adjust the **server_name** and **fullpath** to certificates in each one of these files.

1. In this section you can configure Nginx to bypass the login screen. The auto redirect is enabled goes right to IDP no login button.

```
map $http_x_forwarded_proto $thescheme {  
    default $scheme;  
    https https;  
}  
  
server {  
    listen 80 default_server;  
    server_name netbox.hungry-howard.com;  
  
    return 301 https://$host$request_uri;  
}  
  
server {  
    listen 443 ssl;
```

```

server_name netbox.hungry-howard.com;

ssl_certificate /etc/letsencrypt/live/netbox.hungry-howard.com/fullchain.pem; # managed by Certbot
ssl_certificate_key /etc/letsencrypt/live/netbox.hungry-howard.com/privkey.pem; # managed by Certbot


client_max_body_size 25m;
proxy_set_header X-Forwarded-Host $http_host;
proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-Proto $thescheme;
add_header P3P 'CP="ALL DSP COR PSAa PSDa OUR NOR ONL UNI COM NAV"';
location /static/ {
    alias /opt/netbox/netbox/static;
}

location / {
    proxy_pass http://127.0.0.1:8001;
}

location /login/ {
    proxy_pass http://127.0.0.1:8001/api/plugins/sso/login;
}

location /sso/ {
    proxy_pass http://127.0.0.1:8001/api/plugins/sso/; # Must have a trailing slash to strip the original path
}
}

```

2. Comment out the following lines there will be login Web UI with the SSO button. This is good for testing and to login with the superuser created.

```

map $http_x_forwarded_proto $thescheme {
    default $scheme;
    https https;
}

server {
    listen 80 default_server;
    server_name netbox3.hungry-howard.com;

    return 301 https://$host$request_uri;
}

```

```

server {
    listen 443 ssl;
    server_name netbox3.hungry-howard.com;
    ssl_certificate /etc/letsencrypt/live/netbox3.hungry-howard.com/fullchain.pem; # managed by Certbot
    ssl_certificate_key /etc/letsencrypt/live/netbox3.hungry-howard.com/privkey.pem; # managed by Certbot

    client_max_body_size 25m;
    proxy_set_header X-Forwarded-Host $http_host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-Proto $thescheme;
    add_header P3P 'CP="ALL DSP COR PSAa PSDa OUR NOR ONL UNI COM NAV"';
    location /static/ {
        alias /opt/netbox/netbox/static;
    }

    location / {
        proxy_pass http://127.0.0.1:8001;
    }

    ##### This section is for controlling auto redirect #####
    # location /login/ {
    #     proxy_pass http://127.0.0.1:8001/api/plugins/sso/login/;
    # }

    # location /sso/ {
    #     proxy_pass http://127.0.0.1:8001/api/plugins/sso/; # Must have a trailing slash to strip the original path
    # }
}

```

Configure the setting.py file

Add the following to the bottom of the file.

```

SAML2_AUTH_CONFIG = {
    'AUTHENTICATION_BACKEND': 'django.contrib.auth.backends.RemoteUserBackend',
    'METADATA_AUTO_CONF_URL': "https://global-edjak2.zitadel.cloud/saml/v2/metadata"
}

```

Restart service

```
systemctl restart nginx
```

```
systemctl restart netbox
```

Conclusion:

NetBox configuration.py file should look something like this.

```
ALLOWED_HOSTS = ['netbox.hungry-howard.com']

DATABASE = {
    'NAME': 'netbox',      # Database name
    'USER': 'netbox',      # PostgreSQL username
    'PASSWORD': 'J5brHrAXFLQSif0K', # PostgreSQL password
    'HOST': 'localhost',  # Database server
    'PORT': '',           # Database port (leave blank for default)
    'CONN_MAX_AGE': 300,  # Max database connection age
}

REDIS = {
    'tasks': {
        'HOST': 'localhost',
        'PORT': 6379,
        'USERNAME': '',
        'PASSWORD': '',
        'DATABASE': 0,
        'SSL': False,
    },
    'caching': {
        'HOST': 'localhost',
        'PORT': 6379,
        'USERNAME': '',
        'PASSWORD': '',
        'DATABASE': 1,
        'SSL': False,
    }
}

SECRET_KEY = '!3#qSHaL&(7mw38hdF0kHi98w+6x%N&34t3vh$(Oj^mKZG@$Ax'

ADMINS = [
]

ALLOW_TOKEN_RETRIEVAL = False

AUTH_PASSWORD_VALIDATORS = [
```

```
]
BASE_PATH = ""
CORS_ORIGIN_ALLOW_ALL = False
CORS_ORIGIN_WHITELIST = [
]
CORS_ORIGIN_REGEX_WHITELIST = [
]
CSRF_COOKIE_NAME = 'csrftoken'
DEBUG = False
DEFAULT_LANGUAGE = 'en-us'
EMAIL = {
    'SERVER': 'localhost',
    'PORT': 25,
    'USERNAME': "",
    'PASSWORD': "",
    'USE_SSL': False,
    'USE_TLS': False,
    'TIMEOUT': 10, # seconds
    'FROM_EMAIL': "",
}
ENABLE_LOCALIZATION = False
EXEMPT_VIEW_PERMISSIONS = [
]
INTERNAL_IPS = ('127.0.0.1', '::1')
LOGGING = {
    'version': 1,
    'disable_existing_loggers': False,
    'formatters': {
        'normal': {
            'format': '%(asctime)s %(name)s %(levelname)s: %(message)s'
        },
    },
    'handlers': {
        'file': {
            'level': 'DEBUG',
            'class': 'logging.handlers.WatchedFileHandler',
            'filename': '/var/log/netbox/netbox.log',
            'formatter': 'normal',
        },
    },
}
```

```
'loggers': {
    'django': {
        'handlers': ['file'],
        'level': 'INFO',
    },
    'netbox': {
        'handlers': ['file'],
        'level': 'INFO',
    },
},
}

LOGIN_PERSISTENCE = False
LOGIN_REQUIRED = False
LOGIN_TIMEOUT = None
LOGOUT_REDIRECT_URL = 'home'
METRICS_ENABLED = False
PLUGINS = ['django3_saml2_nbplugin']
PLUGINS_CONFIG = {
    'django3_saml2_nbplugin': {
        'AUTHENTICATION_BACKEND': 'netbox.authentication.RemoteUserBackend',
        'ASSERTION_URL': 'https://netbox.hungry-howard.com/',
        'ENTITY_ID': 'https://netbox.hungry-howard.com',
        'METADATA_LOCAL_FILE_PATH': '/opt/netbox/DCIM.xml',
        'CUSTOM_ATTR_BACKEND': {
            'USERNAME_ATTR': 'emailaddress',
            'MAIL_ATTR': 'emailaddress',
            'FIRST_NAME_ATTR': 'givenname',
            'LAST_NAME_ATTR': 'surname',
            'ALWAYS_UPDATE_USER': True,
        }
    }
}

REMOTE_AUTH_ENABLED = True
REMOTE_AUTH_BACKEND = 'netbox.authentication.RemoteUserBackend'
BANNER_LOGIN = '<a href="/api/plugins/sso/login" class="btn btn-primary btn-block">Login with SSO</a>'
REMOTE_AUTH_AUTO_CREATE_USER = True
RELEASE_CHECK_URL = None
RQ_DEFAULT_TIMEOUT = 300
SESSION_COOKIE_NAME = 'sessionid'
SESSION_FILE_PATH = None
```

```
TIME_ZONE = 'America/Chicago'
DEBUG= True
DATE_FORMAT = 'N j, Y'
SHORT_DATE_FORMAT = 'Y-m-d'
TIME_FORMAT = 'g:i a'
SHORT_TIME_FORMAT = 'H:i:s'
DATETIME_FORMAT = 'N j, Y g:i a'
SHORT_DATETIME_FORMAT = 'Y-m-d H:i'
```

Nginx configuration file

```
map $http_x_forwarded_proto $thescheme {
    default $scheme;
    https https;
}
server {
    listen 80 default_server;
    server_name netbox.hungry-howard.com;
    return 301 https://$host$request_uri;
}
server {
    listen 443 ssl;
    server_name netbox.hungry-howard.com;
    ssl_certificate /etc/letsencrypt/live/netbox.hungry-howard.com/fullchain.pem; # managed by Certbot
    ssl_certificate_key /etc/letsencrypt/live/netbox.hungry-howard.com/privkey.pem; # managed by Certbot
    client_max_body_size 25m;
    proxy_set_header X-Forwarded-Host $http_host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-Proto $thescheme;
    add_header P3P 'CP="ALL DSP COR PSAa PSDa OUR NOR ONL UNI COM NAV"';
    location /static/ {
        alias /opt/netbox/netbox/static/;
    }
    location / {
        proxy_pass http://127.0.0.1:8001;
    }
}
```

Settings.py file

Note: This is a large file so just copied the bottom for example.


```

django_apps = plugin_config.django_apps
if plugin_name in django_apps:
    django_apps.pop(plugin_name)
if plugin_module not in django_apps:
    django_apps.append(plugin_module)
for app in django_apps:
    if "." in app:
        parts = app.split(".")
        spec = importlib.util.find_spec(".".join(parts[:-1]))
    else:
        spec = importlib.util.find_spec(app)
    if spec is None:
        raise ImproperlyConfigured(
            f"Failed to load django_apps specified by plugin {plugin_name}: {django_apps} "
            f"The module {app} cannot be imported. Check that the necessary package has been "
            "installed within the correct Python environment."
        )
INSTALLED_APPS.extend(django_apps)
sorted_apps = reversed(list(dict.fromkeys(reversed(INSTALLED_APPS))))
INSTALLED_APPS = list(sorted_apps)
if plugin_name not in PLUGINS_CONFIG:
    PLUGINS_CONFIG[plugin_name] = {}
plugin_config.validate(PLUGINS_CONFIG[plugin_name], VERSION)
plugin_middleware = plugin_config.middleware
if plugin_middleware and type(plugin_middleware) in (list, tuple):
    MIDDLEWARE.extend(plugin_middleware)
if type(plugin_config.queues) is not list:
    raise ImproperlyConfigured(
        "Plugin {} queues must be a list.".format(plugin_name)
    )
RQ_QUEUES.update({
    f"{plugin_name}.{queue}": RQ_PARAMS for queue in plugin_config.queues
})
SAML2_AUTH_CONFIG = {
    'AUTHENTICATION_BACKEND': 'django.contrib.auth.backends.RemoteUserBackend',
    'METADATA_AUTO_CONF_URL': "https://global-edjak2.zitadel.cloud/saml/v2/metadata"
}

```

root@netbox:/opt/netbox/netbox/netbox#

```
<?xml version="1.0"?>
<md:EntityDescriptor xmlns:md="urn:oasis:names:tc:SAML:2.0:metadata"
    entityID="https://netbox3.hungry-howard.com">
    <md:SPSSODescriptor AuthnRequestsSigned="false" WantAssertionsSigned="false"
protocolSupportEnumeration="urn:oasis:names:tc:SAML:2.0:protocol">
        <md:SingleLogoutService Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-Redirect"
            Location="https://netbox3.hungry-howard.com/" />
        <md:NameIDFormat>urn:oasis:names:tc:SAML:2.0:nameid-format:unspecified</md:NameIDFormat>[]
        <md:AssertionConsumerService Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"
            Location="https://netbox3.hungry-howard.com/api/plugins/sso/acs/"
            index="1" />
    </md:SPSSODescriptor>
</md:EntityDescriptor>
```

Home | NetBox

netbox3.hungry-howard.com/login/?next=/
DjDT

Login with SSO



netbox

Sign In

netbox3.hungry-howard.com (v3.6.5)

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